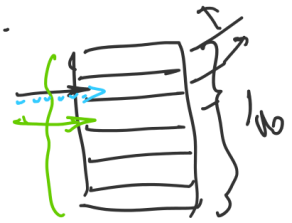
Lecture 5

Saturday, 29 August 2026 11:25 AM

(1) Mini-batch sampling: Training dataset 100 examples, 5 examples in a mini batch select uniformly at random with replacement.



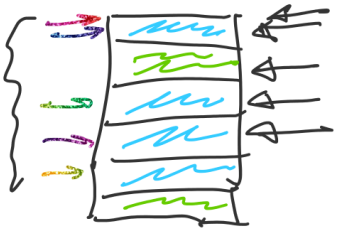
$n = 5$, $m = 100$

$(n_1, n_2 - n_{100})$

$$P_r(\text{all 5 samples are distinct}) = \frac{100}{100} \times \frac{99}{100} \times \frac{98}{100} \times \frac{97}{100} \times \frac{96}{100}$$

$$= \frac{100 \times 99 \times 98 \times 97 \times 96}{(100)^5}$$

Training 100 examples → 5 examples randomly with replacement



60 belong to class A

40 belong to class B

$P(\text{mini-batch containing at least one example from both classes}) \Rightarrow P(\varepsilon) = 1 - P(\varepsilon^c)$

$P(\varepsilon^c) = P(\text{all examples from A or all examples being from class B})$

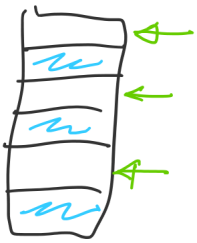
$$\left(\frac{3}{5}\right)^5 + \left(\frac{2}{5}\right)^5 = P(\text{all examples class A}) + P(\text{All examples from class B})$$

$$\left(\frac{40}{100}\right)^5 = \left(\frac{2}{5}\right)^5$$

$$\frac{60}{100} \times \frac{60}{100} \times \frac{60}{100} \times \frac{60}{100} \times \frac{60}{100} = \left(\frac{60}{100}\right)^5 = \left(\frac{3}{5}\right)^5$$

Dataset Quality check

Image dataset → 1000 labeled images
→ 50 images are wrongly labeled.



Sample 20 distinct images from the dataset & I check whether each of them is correctly labeled

$$P(\text{QC did not find any incorrectly labeled images}) = \frac{\binom{950}{20}}{\binom{1000}{20}} \cdot \binom{50}{20}$$

$$= \frac{1500!}{201!\,980!}$$

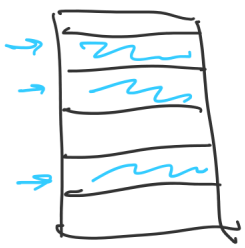
$$P(\text{At least one incorrect image is selected}) = 1 - P(\text{All correct images are selected})$$

$$= 1 - \frac{\binom{950}{20}}{\binom{1000}{20}}$$

③ $P(\text{Selected 2 incorrectly labeled images})$

$$= \frac{\binom{50}{2} \cdot \binom{950}{18}}{\binom{1000}{20}}$$

$$P(X \geq 2) = 1 - (P(X=0) + P(X=1))$$



Eg → Classification Errors.

On each test example, it classifies incorrectly w.p. $.2$. Independent errors.

10 examples

① P(classifier makes exactly 3 errors) ✓

$\binom{10}{3}$ ~ C C C C C C C E E E ✓ ①

~ C E E C C C C C C E E ✓ ②

③

$\left\{\binom{10}{3}\right.$ $(.8)(.8)(.8)\cdots(.8)(.2)(.2)(.2)$ ✓

~ $(.8)(.2)(.2)(.8)\cdots(.8)(.2)(.8)$

✓ $\boxed{\binom{10}{3}(.8)^7 \times (.2)^3}$ ✓ $\binom{10}{3} = \binom{10}{7}$

Binomial probabilities

n ind. exp. chance of success in each experiment $= p$

P(k successes in n exp.)

$$= \binom{n}{k} \cdot p^k (1-p)^{n-k}$$

$\binom{10}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^7$

$\binom{10}{3}\frac{1}{2^{10}}$



class A = 60

class B = 40

10 examples from the dataset

P(success) = $\frac{6}{10}$

{ P(collect 4 samples from class A)

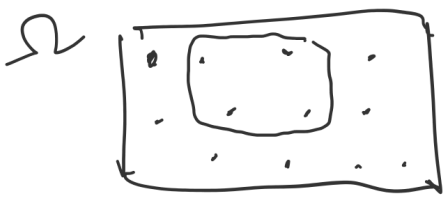
n → 10 samples . Success - class A

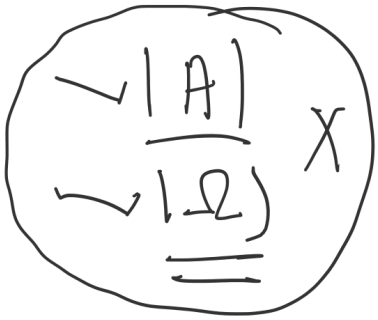
$(\cdot 8)^7 \times (\cdot 2)^3$

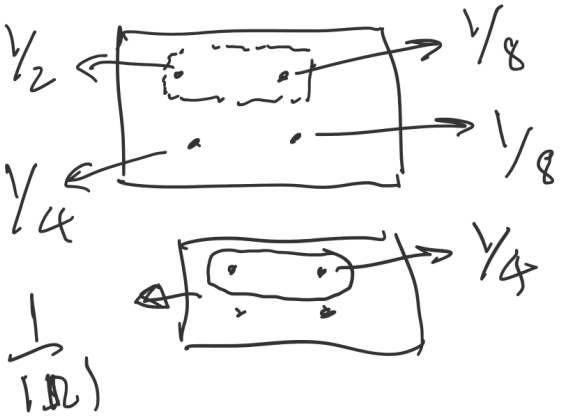
C C C C C E E E C C

E C C C C C C C C C

$(\cdot 8)^9 \times (\cdot 2)$







$P(A) = \frac{2}{4}$ ✗

$P(A) = \frac{1}{2} + \frac{1}{8}$

$$\sum_{K=0}^{n} \binom{n}{K} p^K (1-p)^{n-K} = 1$$

$$(p + 1 - p)^n$$

C E C

P(3 errors | 2nd image being wrongly classified)

$= \dfrac{P(\text{3 errors} \cap \text{2nd image wrongly})}{P(\text{2nd image wrongly classified})}$

$(\cdot 2)$

ε

$(\cdot 8)^2(\cdot 2)^2$ ↘ C C E E → A_1 — 4 images, 2 errors

$\hookrightarrow 2 \times (\cdot 8)^2 \times \cdot 2$ ↘ E C C E → A_2

A ↘ E E C C → A_3 — P(2 errors in 4 images)

↘ E C E C ⋮ $= P$(Config gen. lies in A)

↘ C E C E ⋮ $= \sum_{i=1}^{|A|} P$(Config gen is A_i)

↳ E E E C

$= |A| \cdot (\cdot 8)^2 (\cdot 2)^2$